Explicit Transformations from Edge-Depth-Robust to Node-Depth-Robust Graphs with Improved Concrete Efficiency

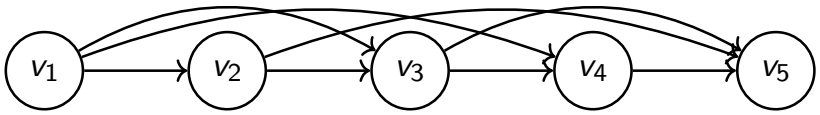
Jeremiah Blocki Nathan Smearsoll

Purdue University, West Lafayette, USA
jblocki@purdue.edu, nsmearso@purdue.edu

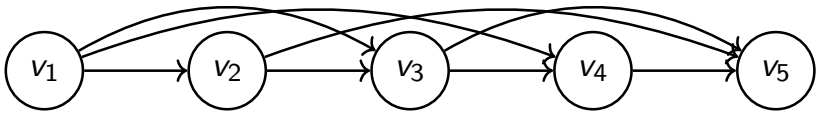
November 2026

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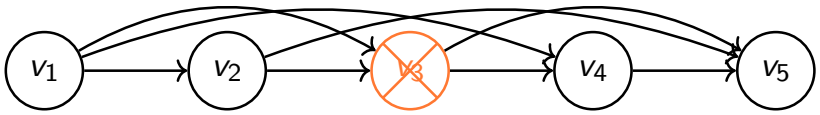
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Node Depth-Reducibility

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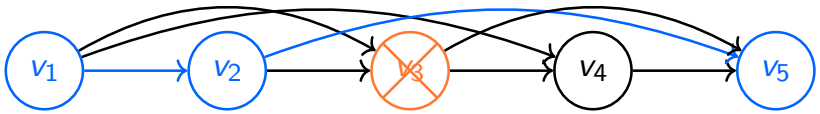
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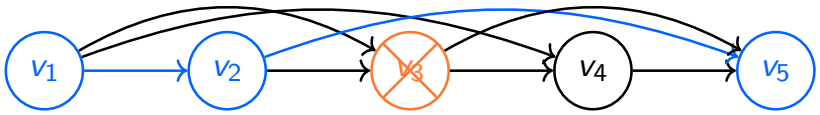
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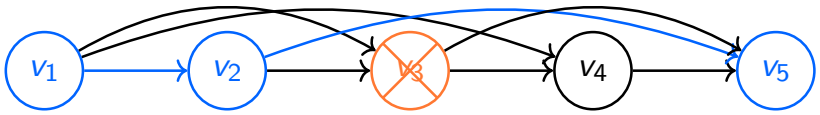
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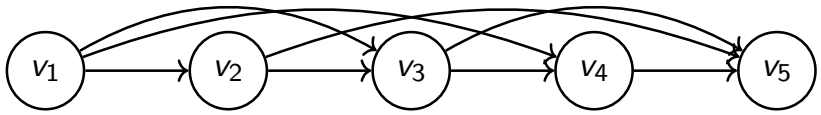
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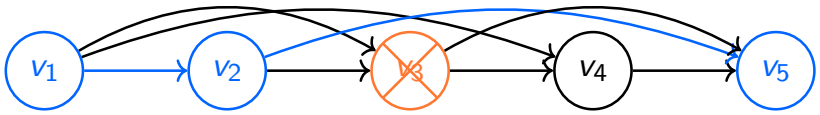
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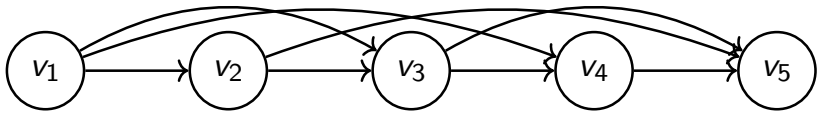
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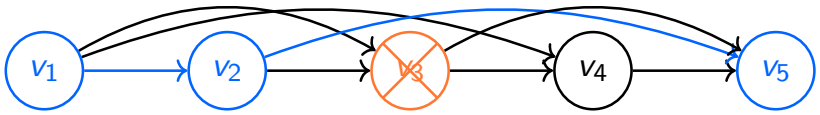
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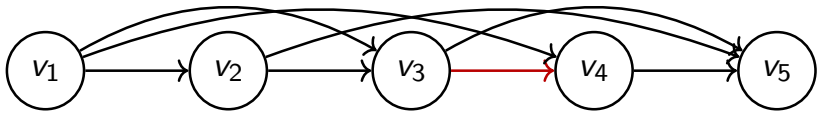
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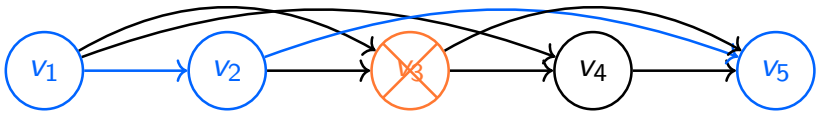
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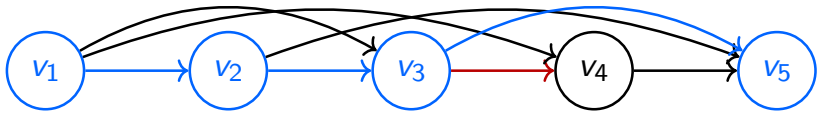
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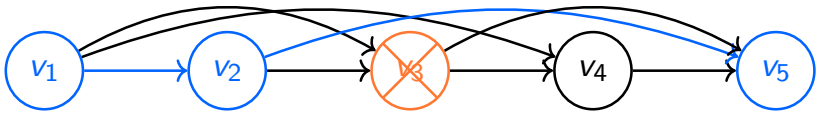
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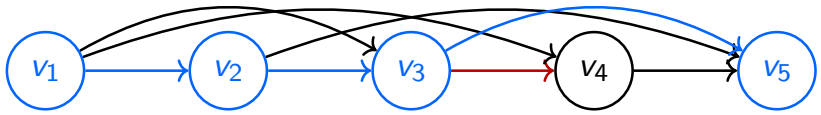
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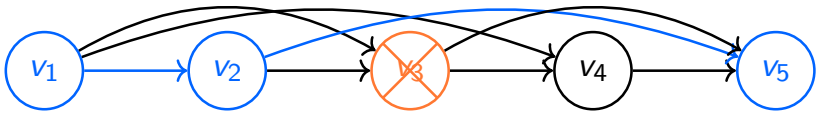
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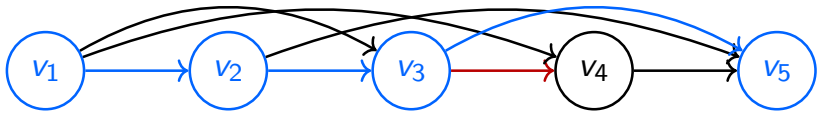
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Potential Solution

Can we transform an edge depth-robust graph into a node depth-robust graph?

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2. Not explicit.
3. Increases number of nodes — $N' = O(M)$.

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Theorem (Special Case)

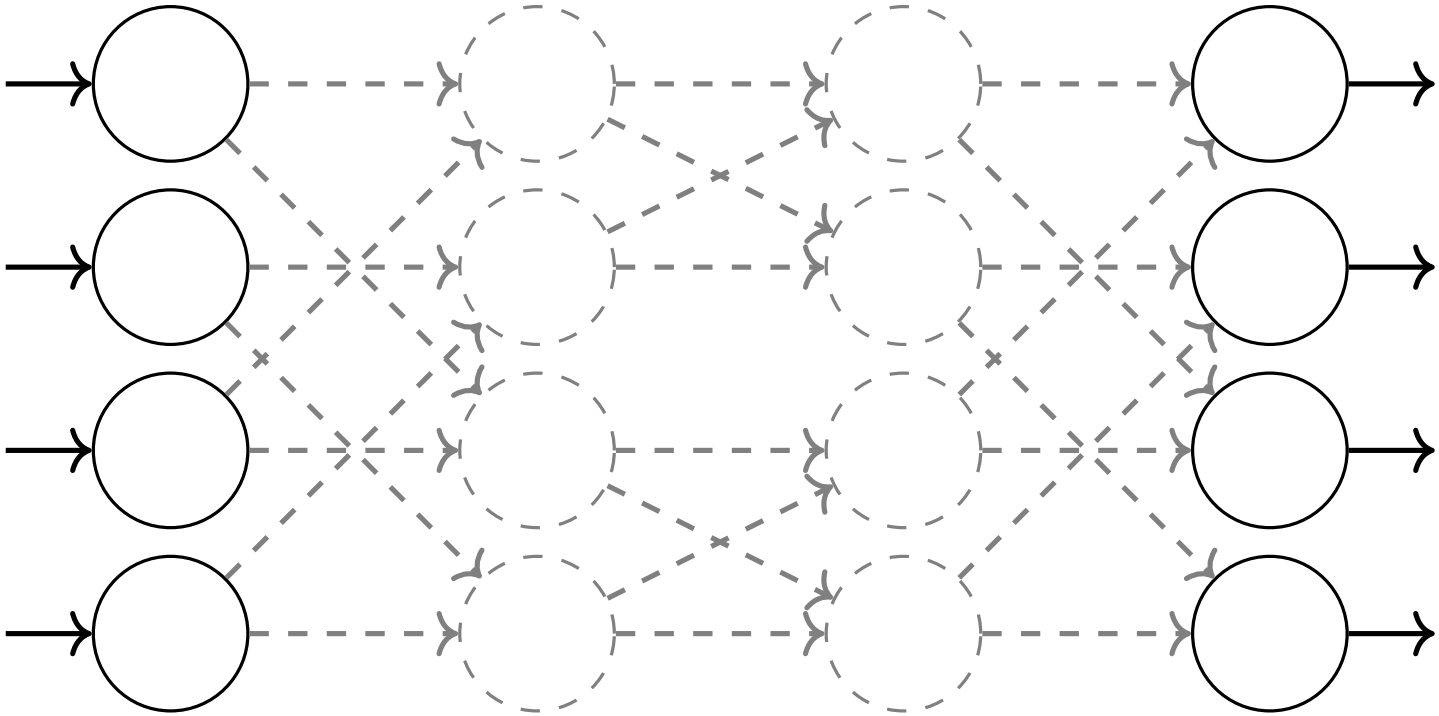
Given a graph G_N with $2^{n+1} - 1$ nodes and $n2^n$ edges, which is (e, d)-edge depth robust.
$\textsc{Transform}(G_N, \mathrm{SC}^{depth}, I, O)$ *is a graph with $O(n2^n)$ nodes and indegree 2.*
$\textsc{Transform}(G_N, \mathrm{SC}^{depth}, I, O)$ *is $(e/3, (2d - 1 - e) \cdot n)$-node depth robust for all $d > 1$.*

Our Transformation

Key tool: Superconcentrators.

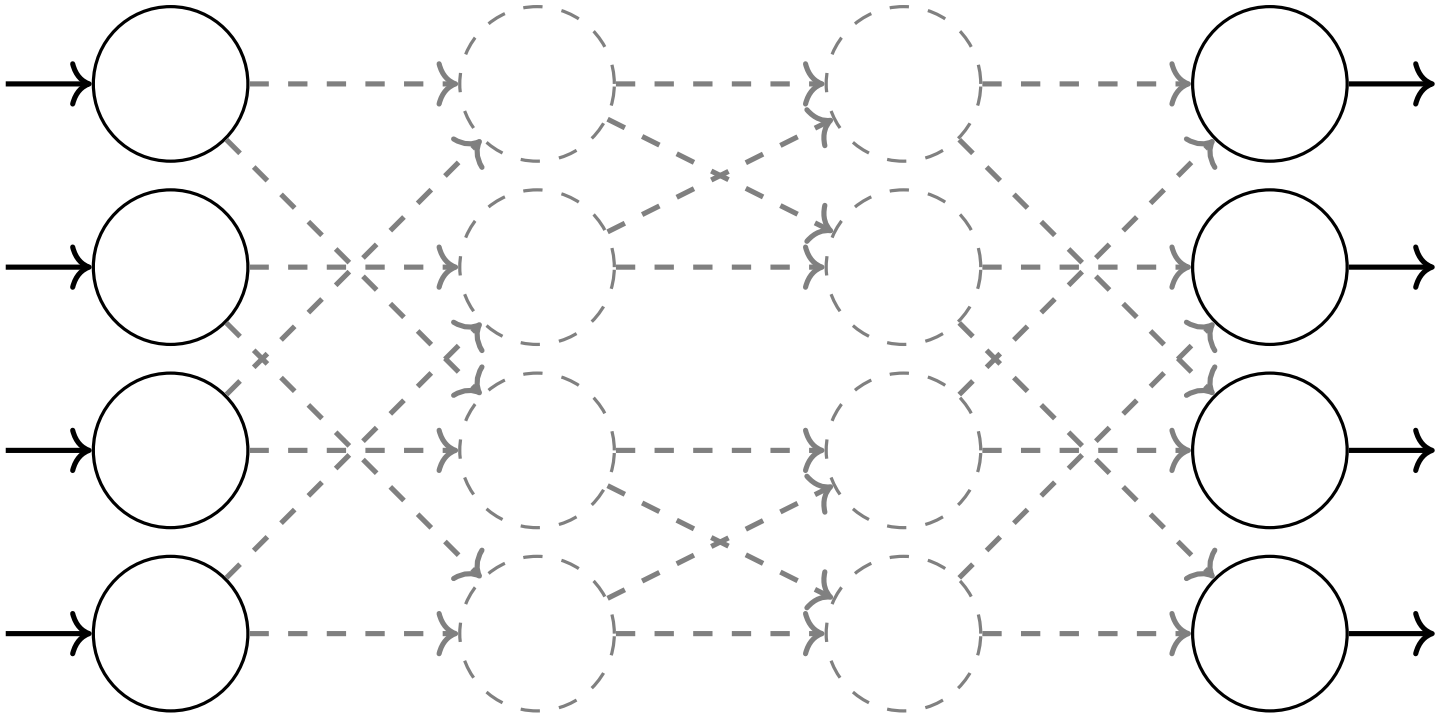
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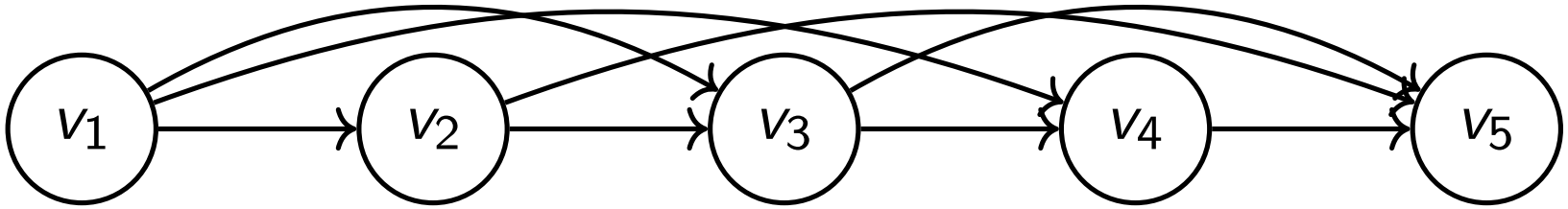
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1. n inputs and outputs.
2. Strong connectivity properties, even after node deletions.
3. Prior works give linear-sized constructions — including explicit ones.

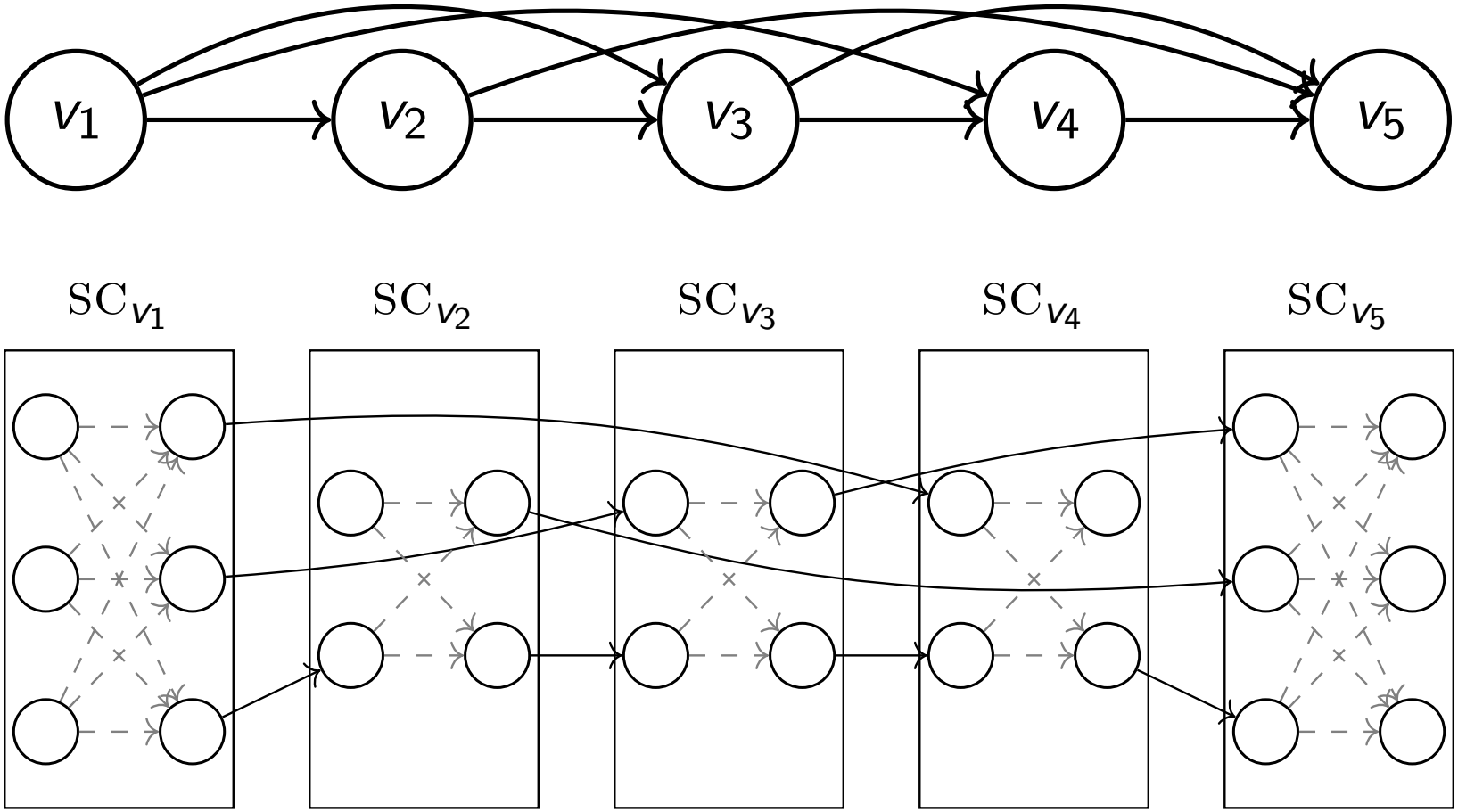
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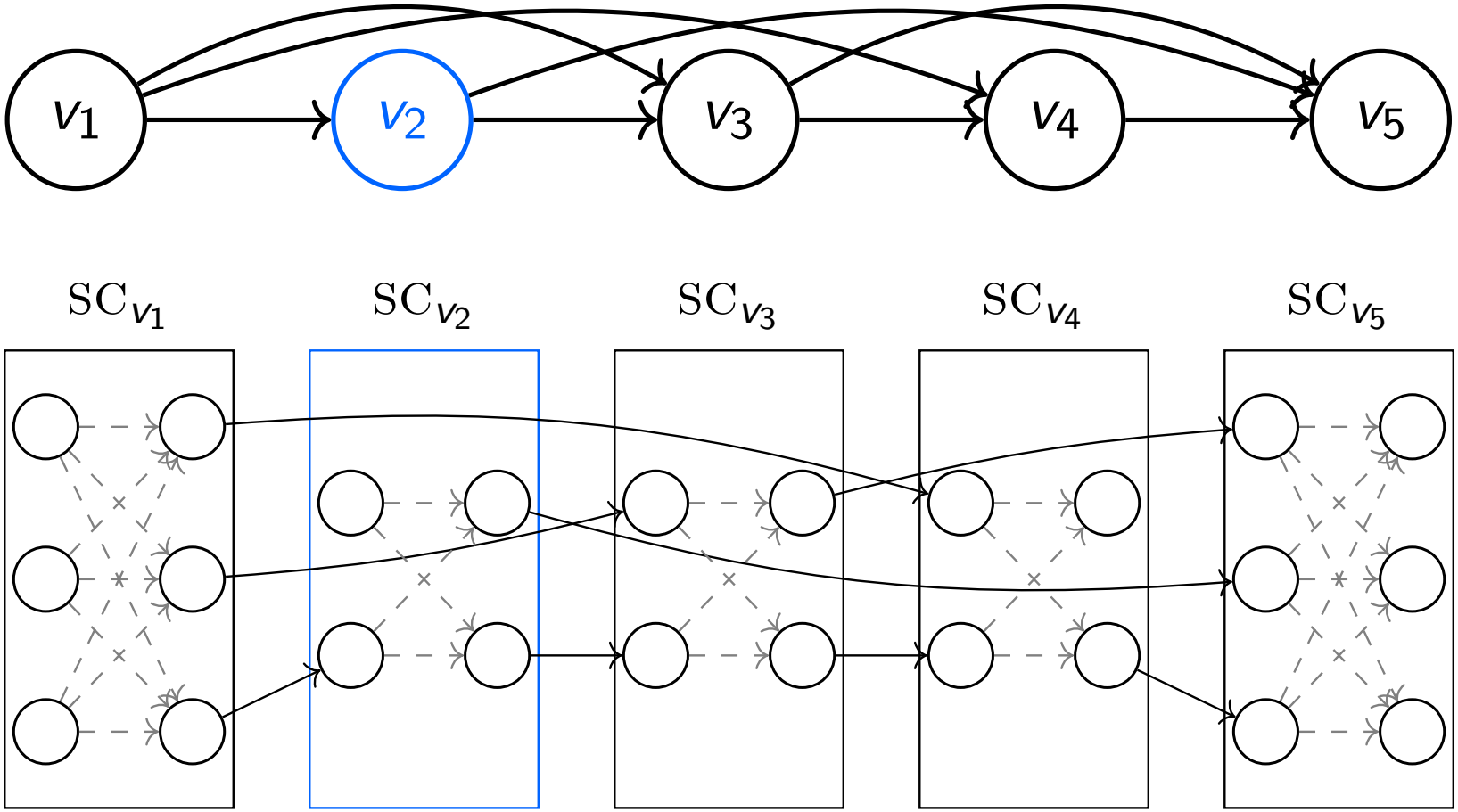
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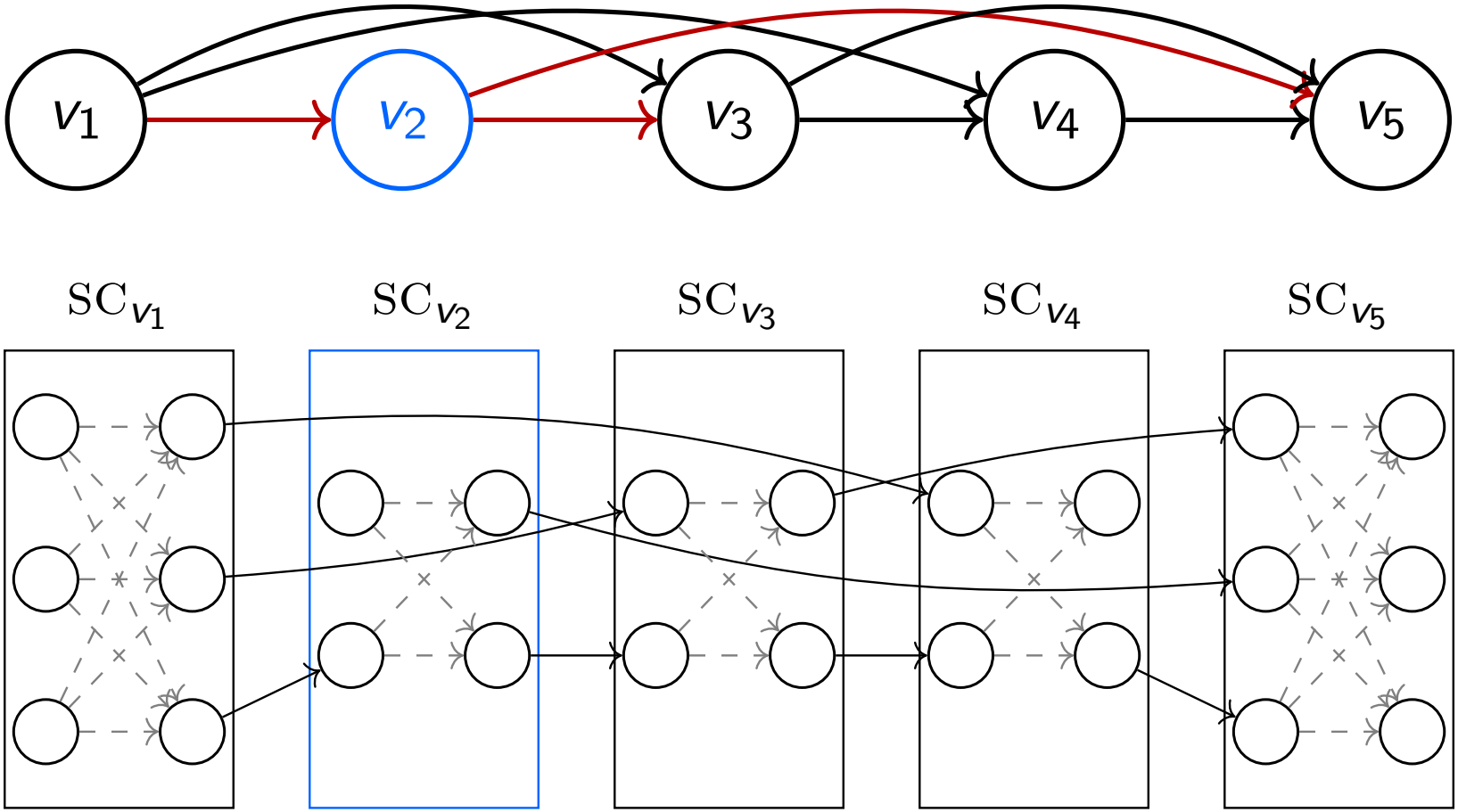
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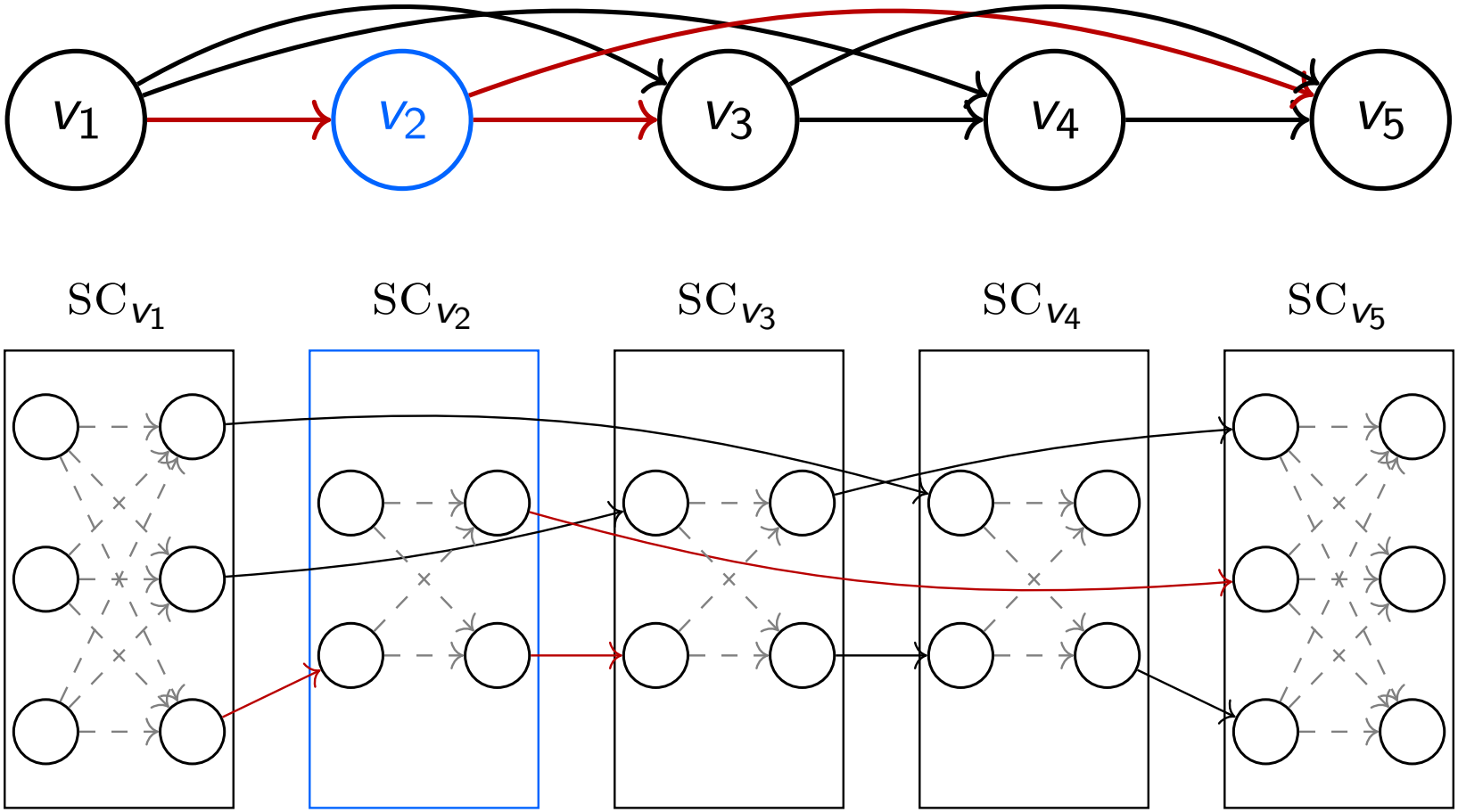
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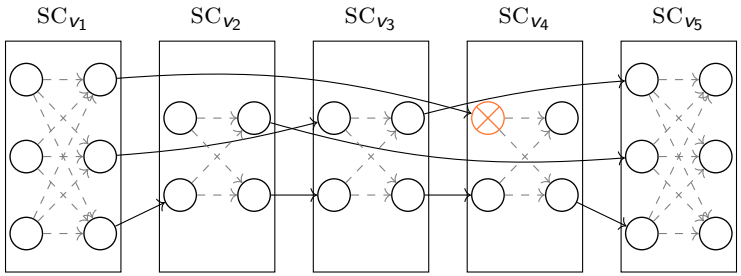
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We map *any* subset $S^* \subseteq V(G_N^*)$ to a corresponding set $S \subseteq E(G_N)$, and show that $|S| \leq |3S^*|$.

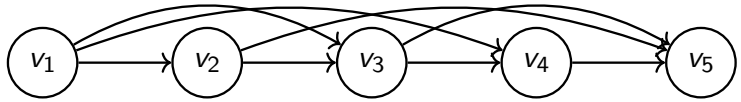
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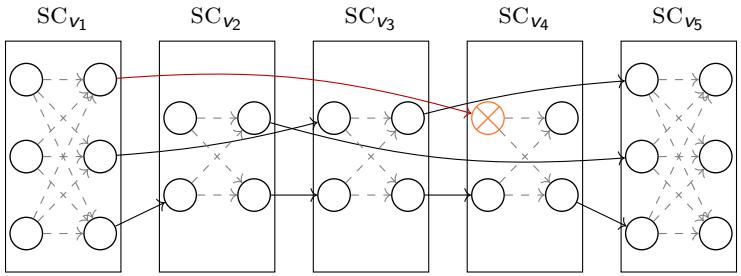
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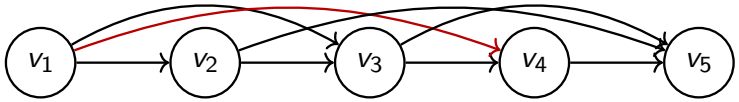
Proving Our Transformation Preserves Robustness

We show that $G_N^* = \textsc{Transform}(G_N, \mathrm{SC}, I, O)$ is node-Depth Robust as long as G_N is edge-Depth Robust.

$S^* \subseteq V(G_N^*)$



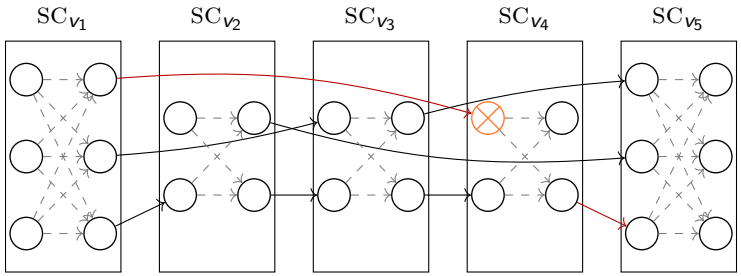
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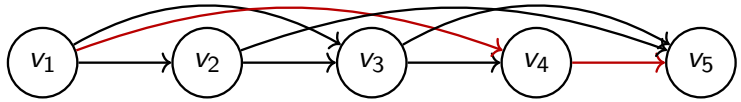
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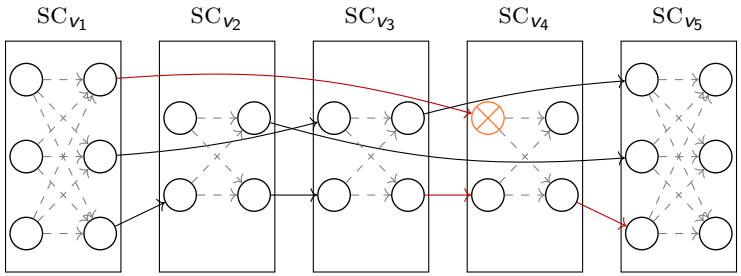
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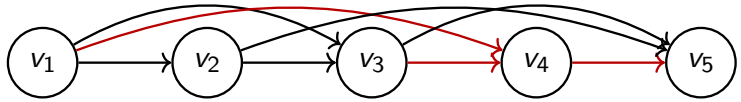
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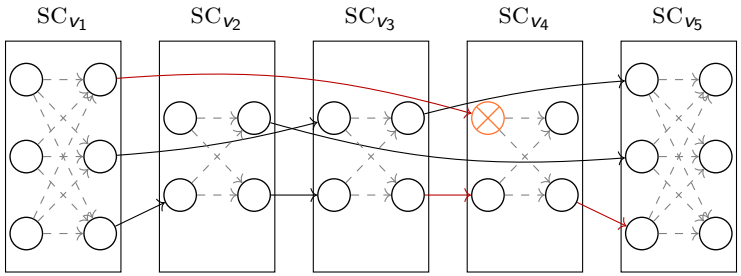
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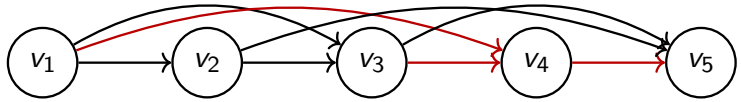


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$S^* \subseteq V(G_N^*)$ $\qquad$ $S \subseteq E(G_N)$





We split the impact of removing S^* into three sets of removed edges: $\textsc{Dest}$, $\textsc{IntDisr}$, and $\textsc{ExtDisr}$. Each has a corresponding set of removed edges in G_N.

Proving Our Transformation Preserves Robustness

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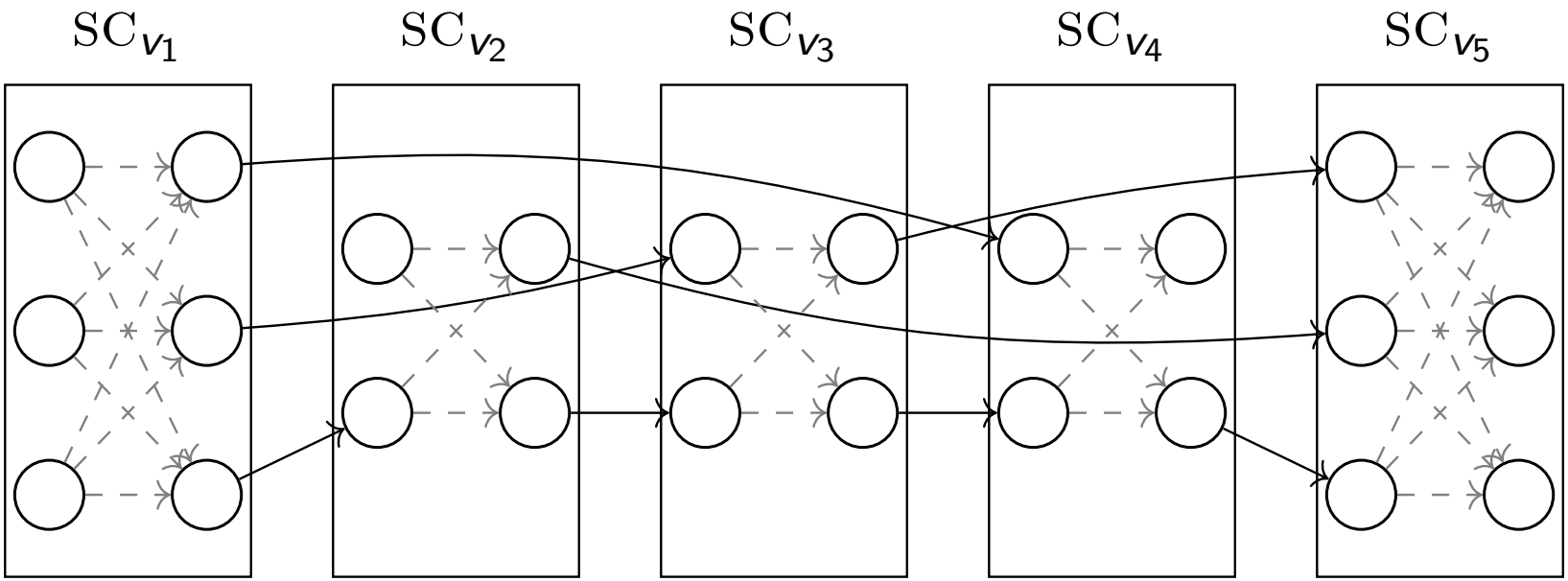
Definition (Destroyed Edges)

DEST is the set of all edges whose destination node is directly removed.

Proving Our Transformation Preserves Robustness

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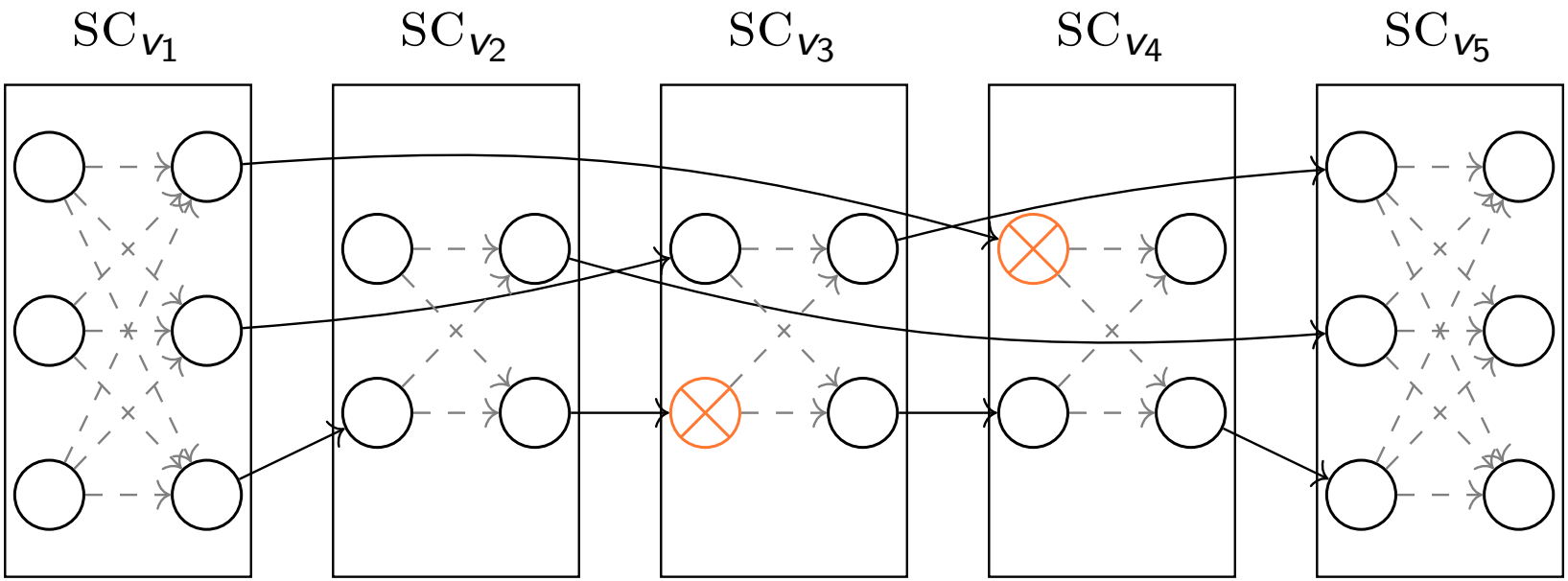
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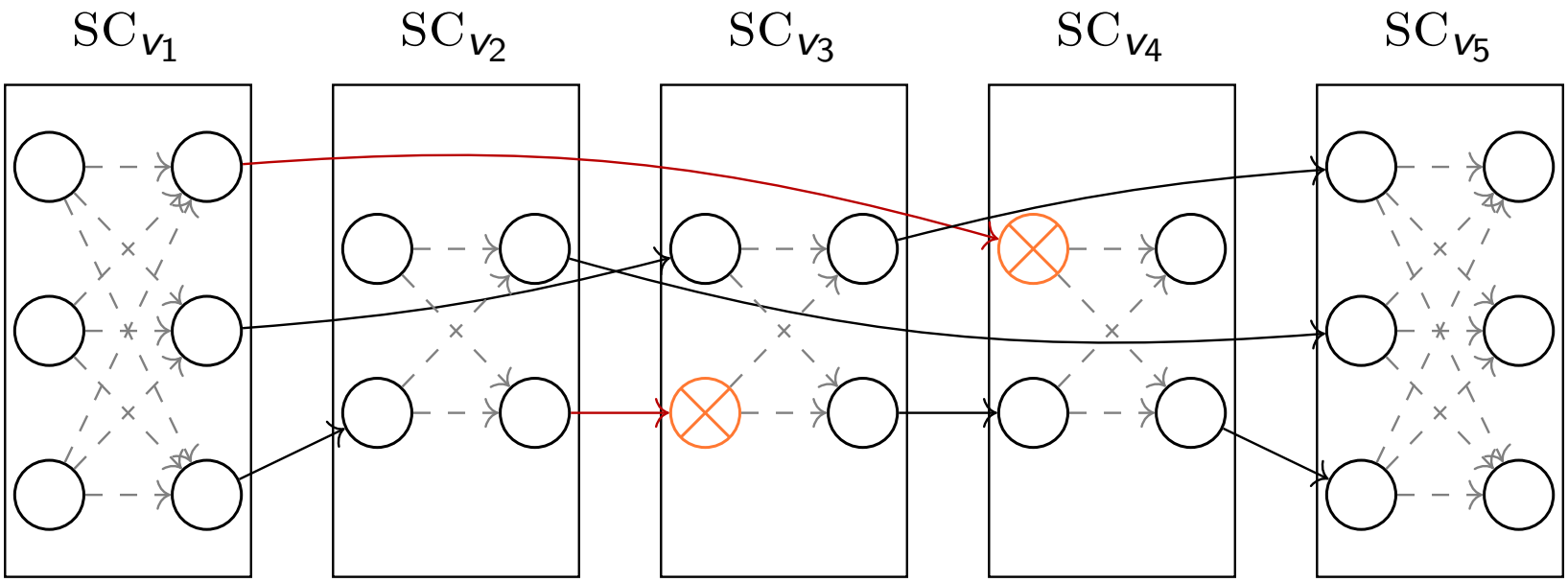
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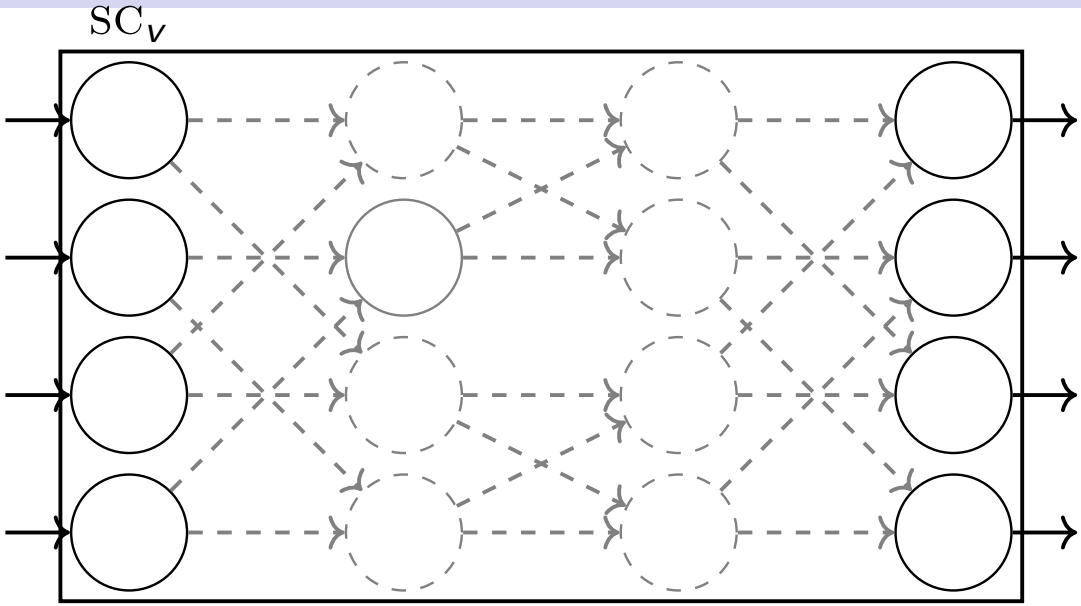
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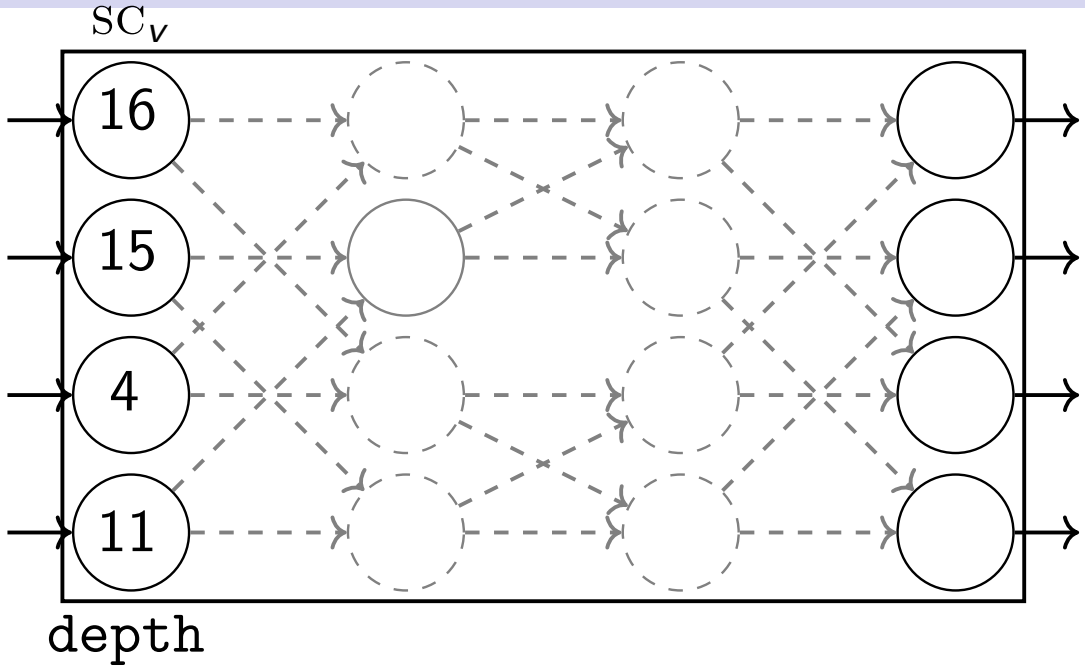


We trivially bound $|S_{\mathrm{DEST}}| \leq |S^*|$.

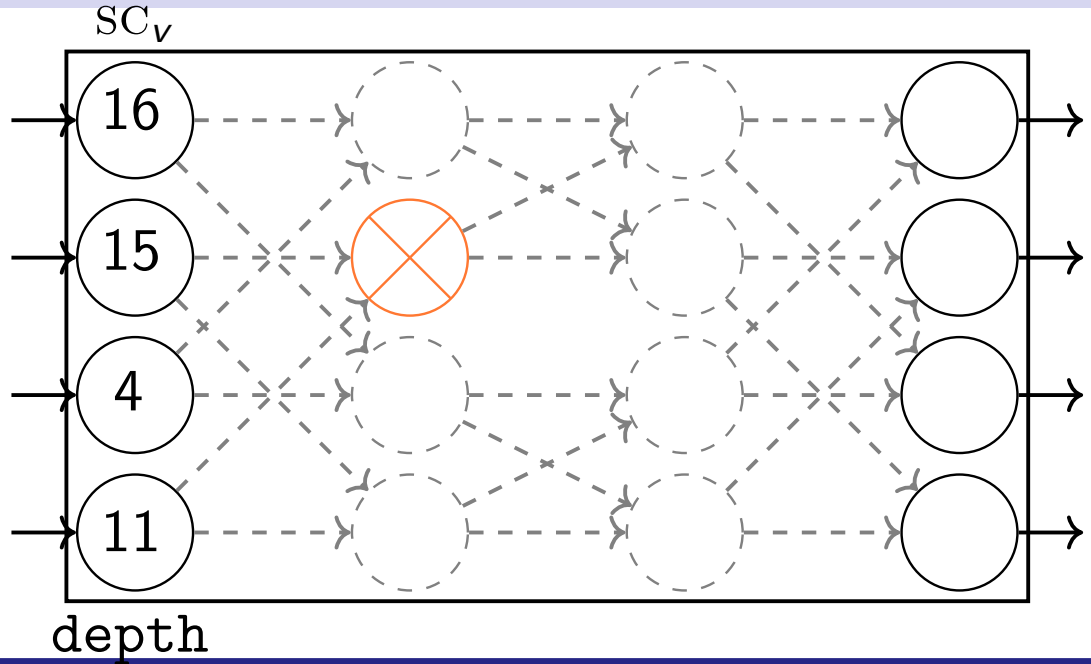
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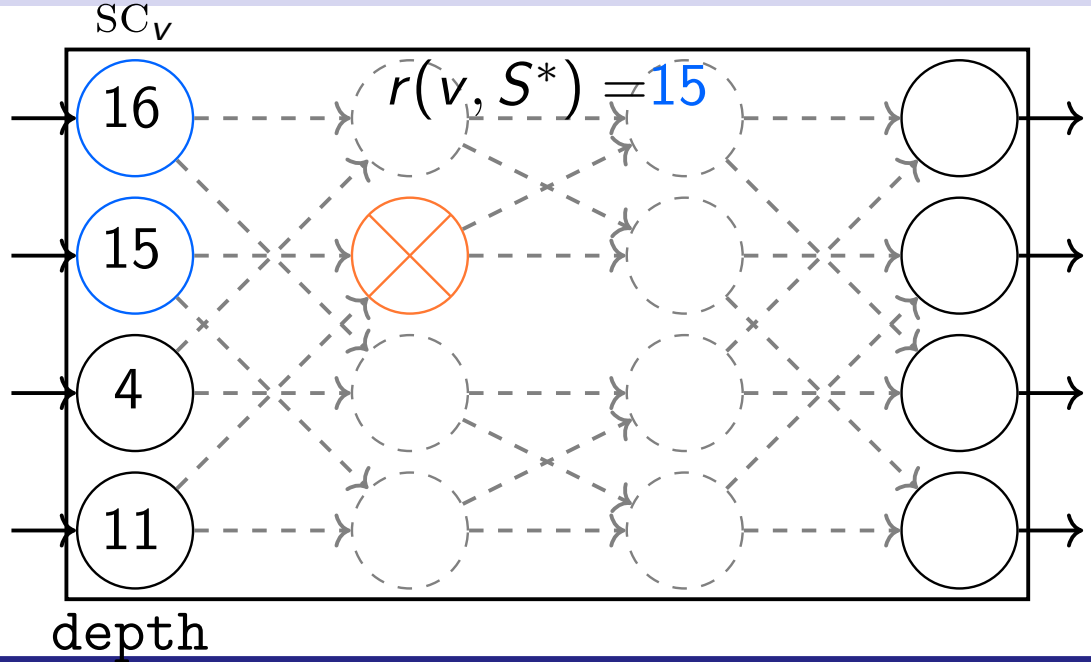


Definition (r)

$$r(v, S^*) \doteq \max\{t : |\{x \in \mathrm{IN}(\mathrm{SC}_v) \ : \ \mathtt{depth}(x, G_N^* - S^*) \geq t\}| > |S_v^*|\}$$

Intuitively, this is the largest depth t such that we have more nodes of that depth than removed nodes.

Proving Our Transformation Preserves Robustness

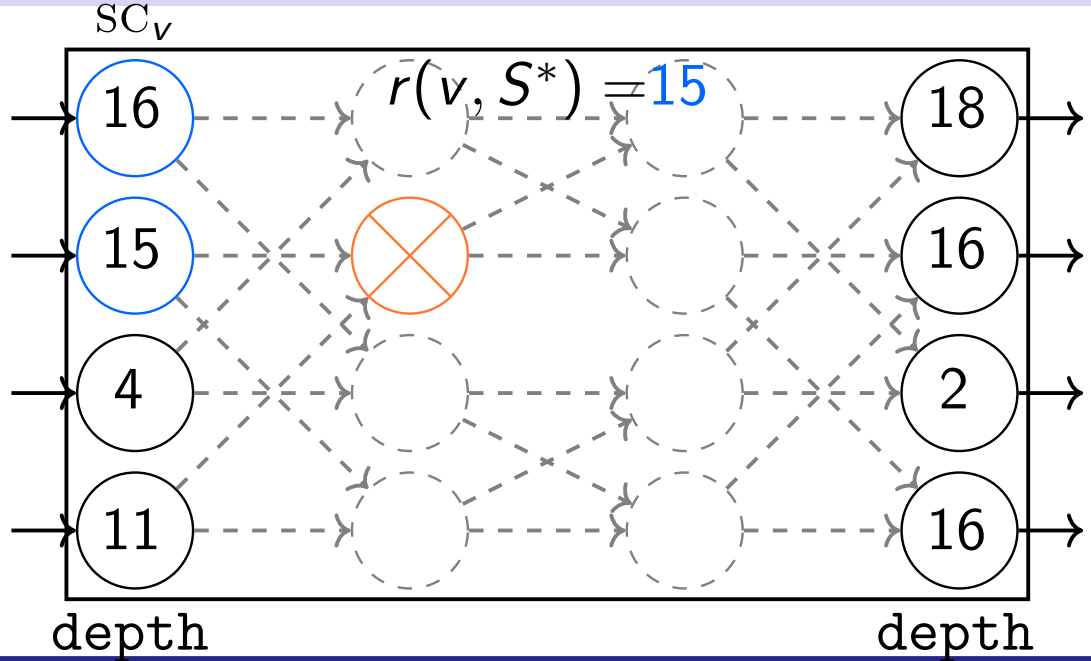


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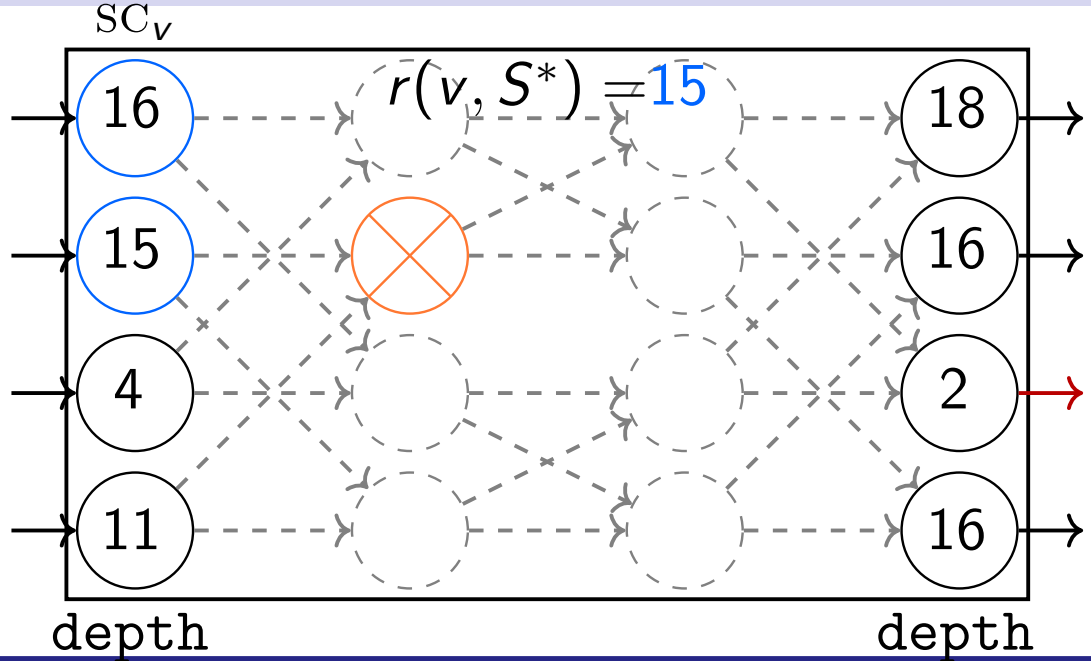
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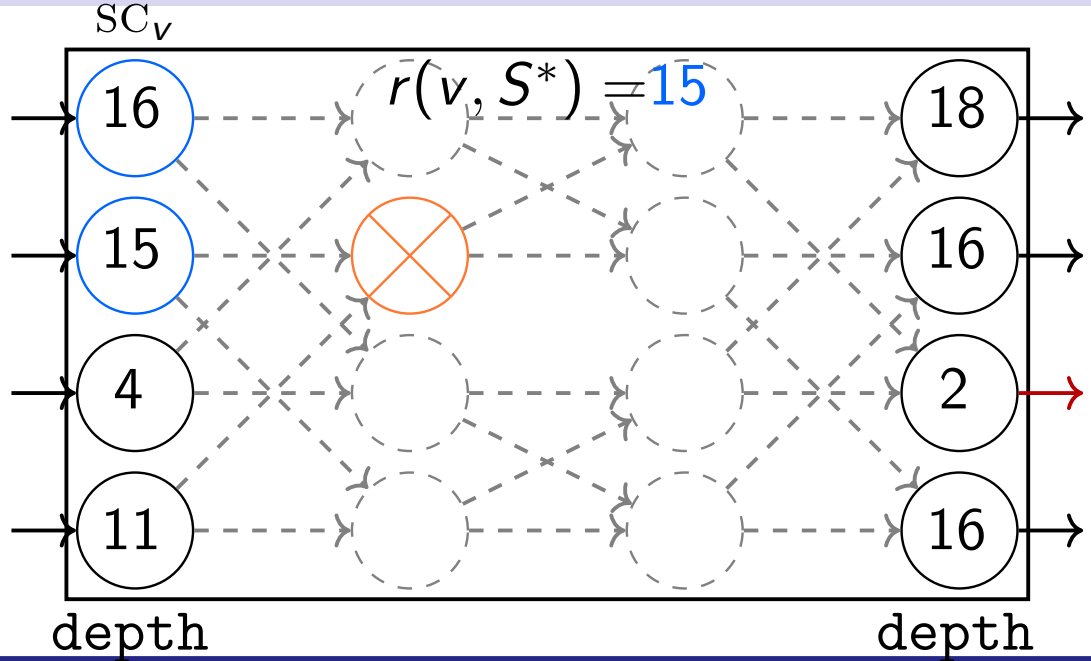
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Definition (Internally Disrespectful Edges)

IntDisr is the set of all edges (a, b) with a source a that both (1) is an output of some SC_v and (2) has insufficient $\mathtt{depth}(a) < r(v, S^*) + 1$.

Proving Our Transformation Preserves Robustness



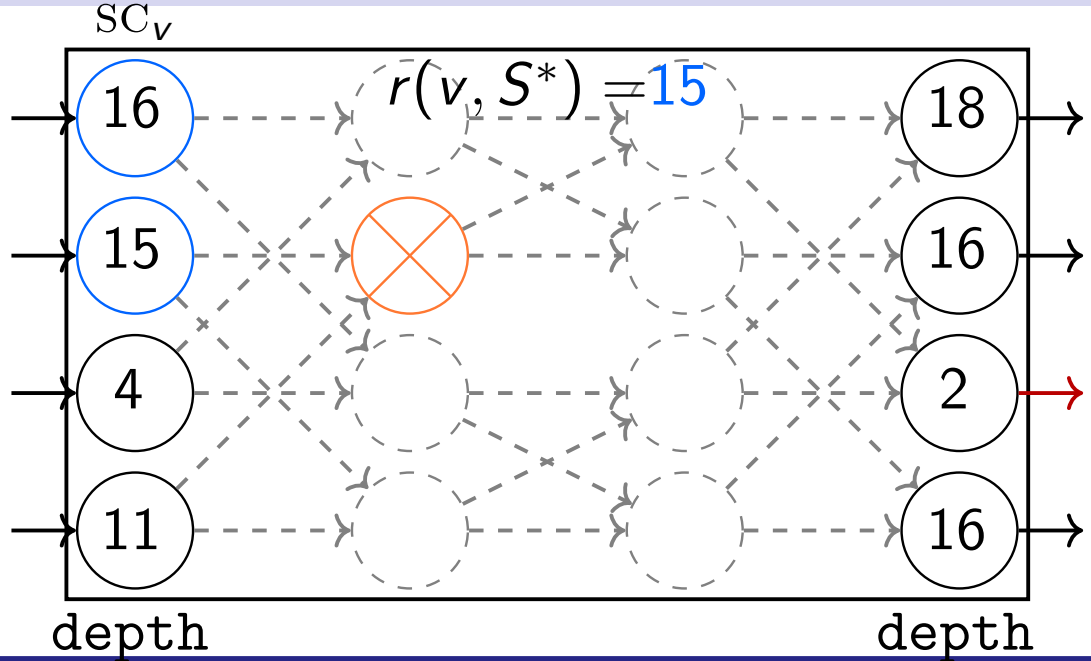
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We bound $|S_{\mathrm{IntDisr}}| \leq |S^*|$ by appealing to superconcentrator properties.

Proving Our Transformation Preserves Robustness

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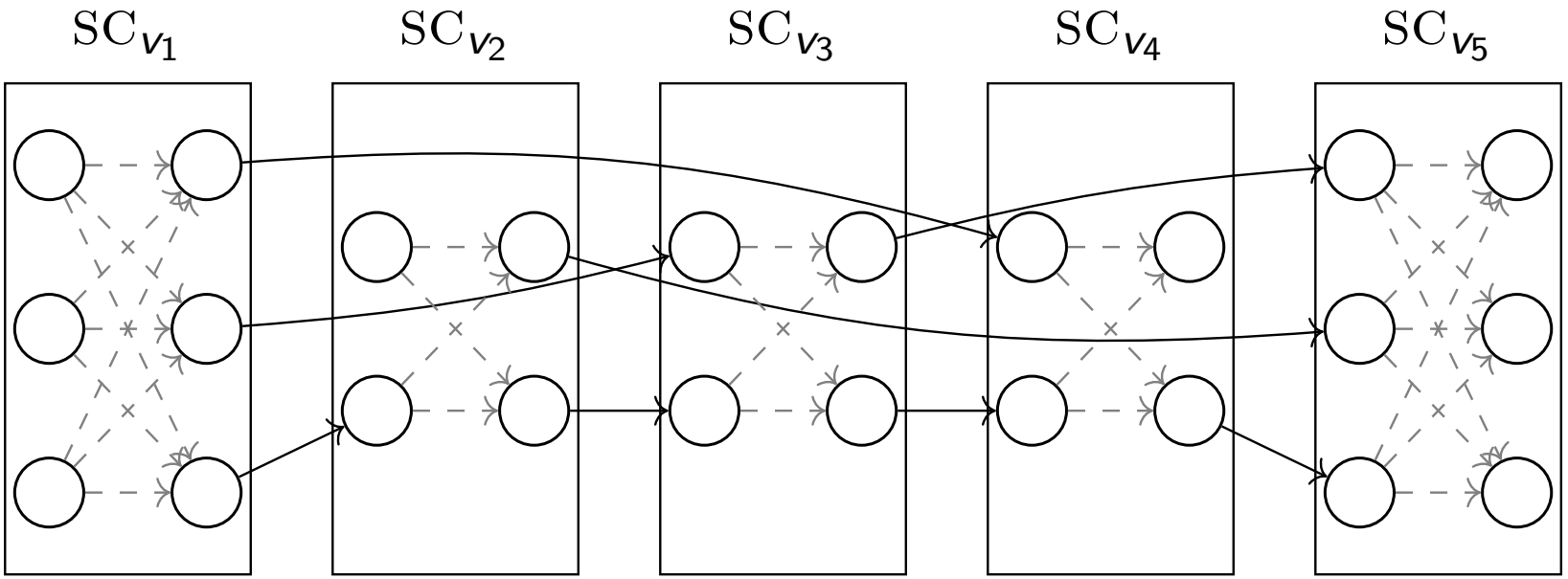
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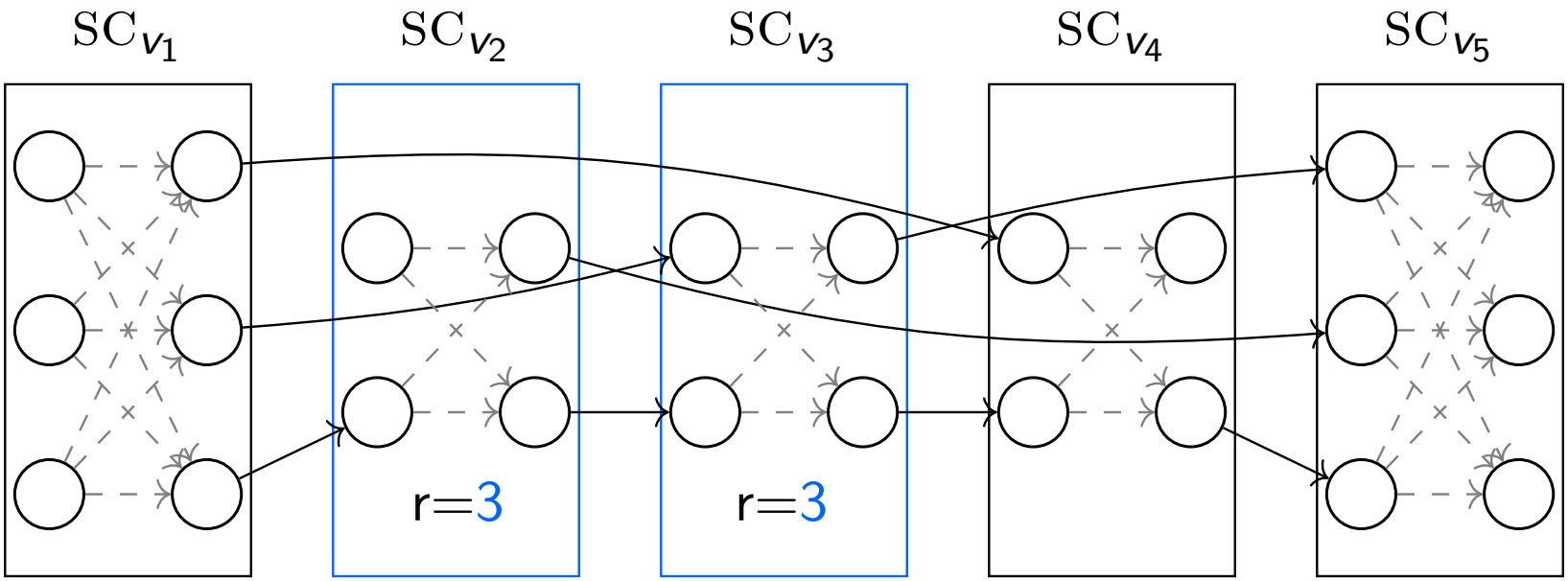
What if the r values do not increase in the order we expect?



Proving Our Transformation Preserves Robustness

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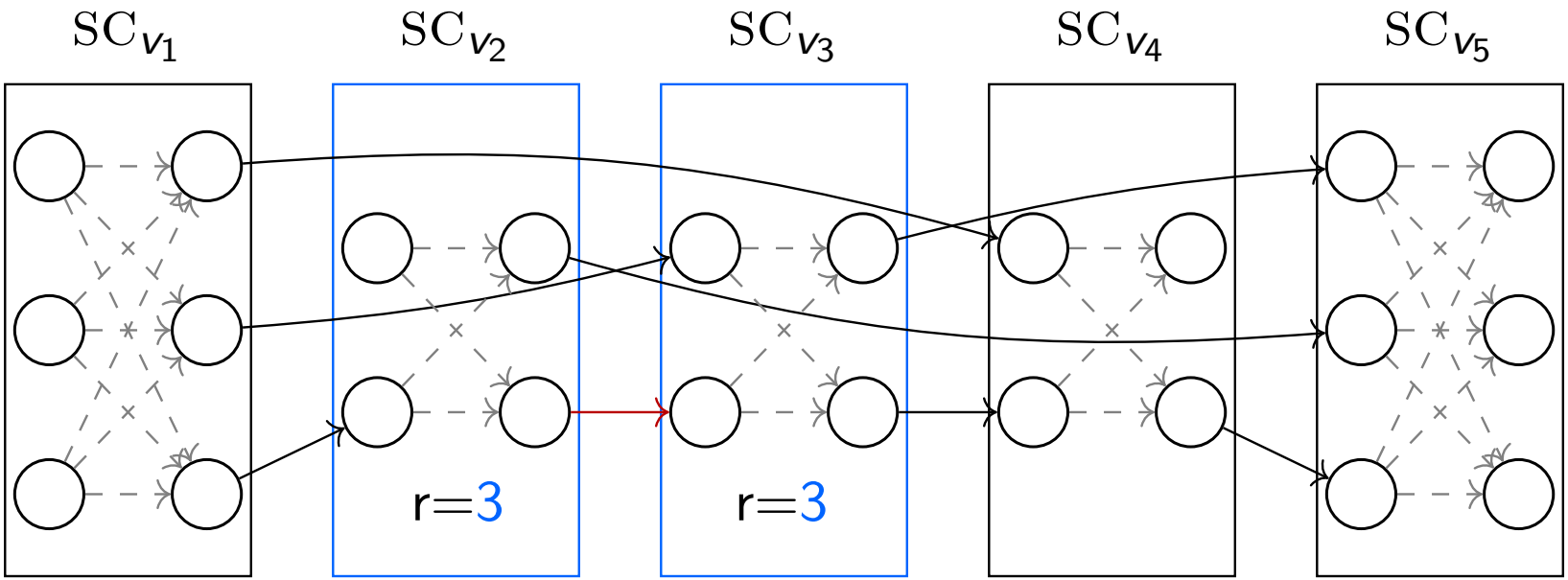
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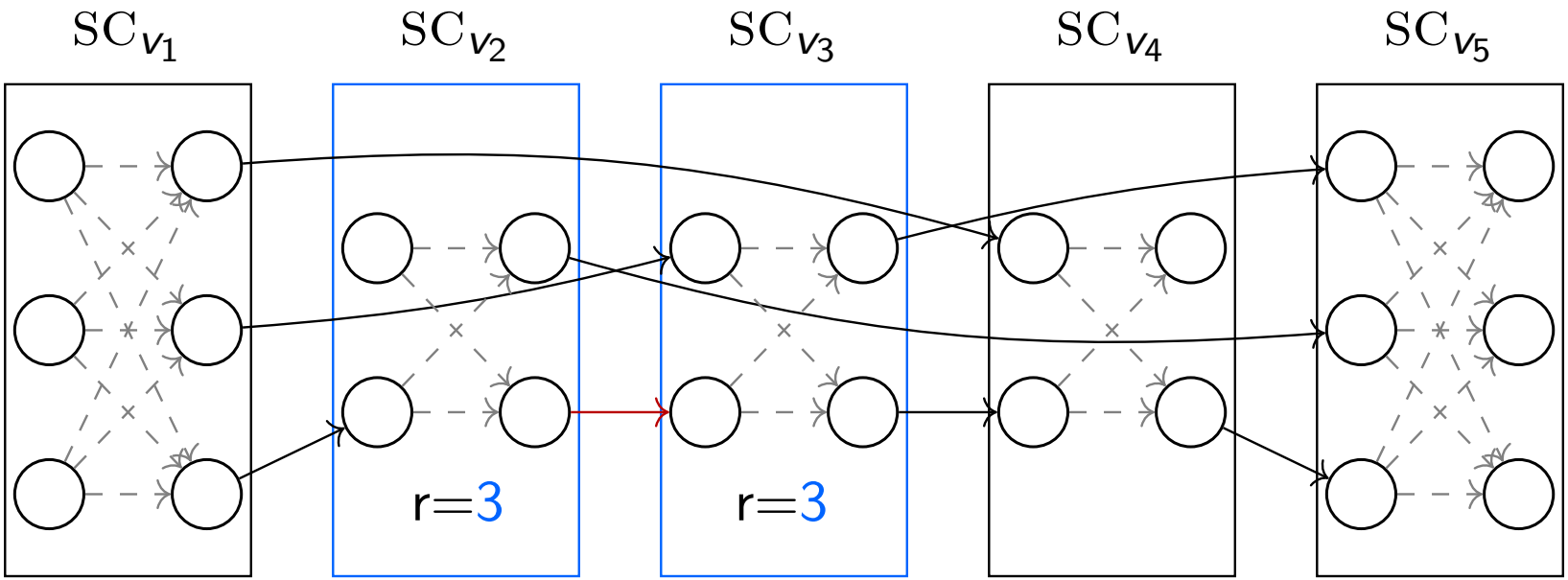
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We add all edges (u, v) with $r(v) \leq r(u) + 1$ to ExtDisr.

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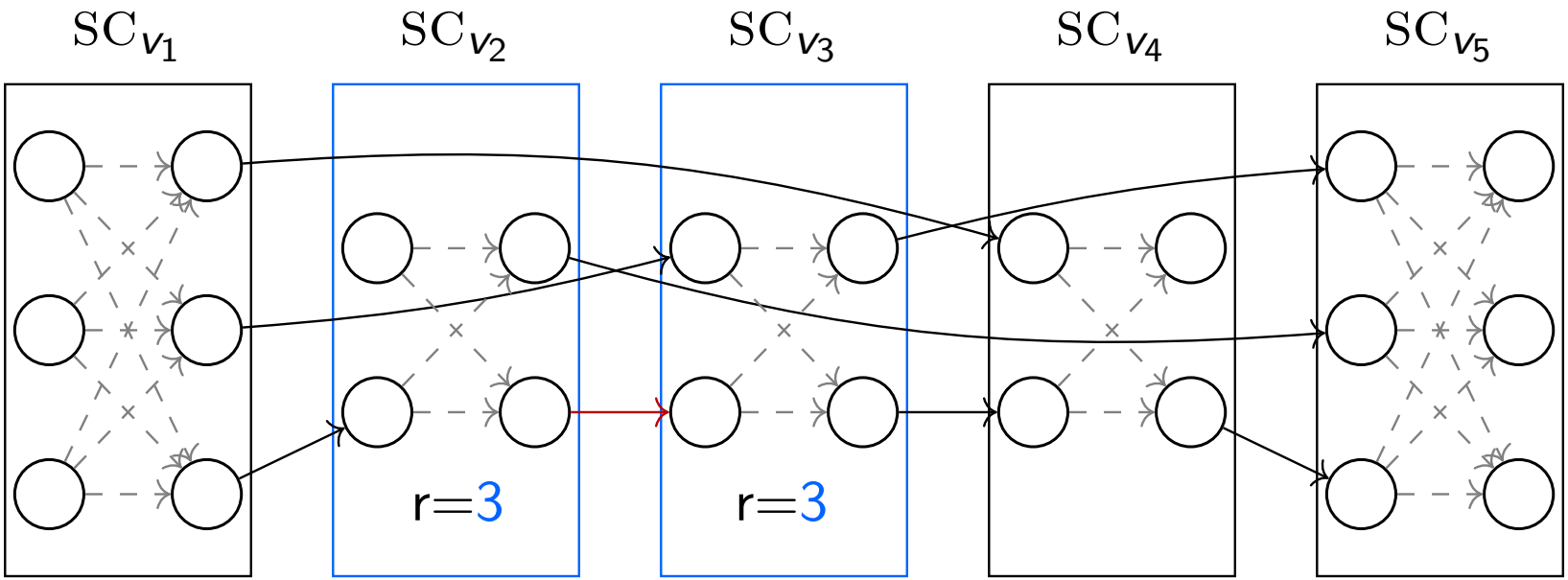
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We bound $|S_{\text{ExtDisr}}| \leq |S^*|$.

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Let $S = \textsc{Dest} \cup \textsc{IntDisr} \cup \textsc{ExtDisr}$. We know $|S| \leq 3|S^*|$. We argue that any path in $G_N - S$ corresponds to a series of increasing r values in $G_N^* - S^*$. Thus, $G_N^* = \textsc{Transform}(G_N, \text{SC}, I, O)$ is $(e/3, 2d-1)$ node-Depth Robust as long as G_N is (e, d) edge-Depth Robust.

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Concrete Analysis of Our Results

We also provide a novel analysis of Grates showing it is $(e, 2d-1)$-*edge* depth-robust for $e + d \leq 2^n$. We apply TRANSFORM to Grates, and show that TRANSFORM(*Grates*) is $(e/3, (2d - e - 1)n)$ *node* depth-robust*

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